BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

BIOLOGY

Past Paper 2024 • Paper Code: 6463 / 6464

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Biology (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6463

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Production of glucose is most closely associated with:

(A) Aerobic respiration
(B) Anaerobic respiration
(C) Krebs cycle
(D) Calvin cycle

Q2. Which group would you assign to a plant which produces spores and embryo but lack seed and vascular tissue?

(A) Angiosperm
(B) Bryophyte
(C) Algae
(D) Gymnosperm

Q3. Normal pH of human blood is:

(A) 6.4
(B) 7.4
(C) 8.4
(D) 4.7

Q4. The systolic pressure of normal human is:

(A) 70 mm Hg
(B) 90 mm Hg
(C) 100 mm Hg
(D) 120 mm Hg

Q5. Spiracles are found in:

(A) Fishes
(B) Cockroach
(C) Frog
(D) Birds

Q6. Dipeptides are broken down into amino acid by an enzyme called:

(A) Erypsin
(B) Trypsin
(C) Lactase
(D) Maltose

Q7. Pyruvic acid is produced as result of:

(A) Calvin cycle
(B) Glycolysis
(C) Electron transport chain
(D) Krebs cycle

Q8. Syrinx is organ of voice of:

(A) Goat
(B) Crow
(C) Toad
(D) Snake

Q9. Commercially shark-liver oil is extracted and used as source of vitamin:

(A) B
(B) C
(C) B_{12}
(D) A & D

Q10. Loose smut of wheat is caused by:

(A) Phytophthora
(B) Rhizopus
(C) Puccinia
(D) Ustilago

Q11. Kelps are largest of known algae which belongs to:

(A) Diatoms
(B) Red algae
(C) Brown algae
(D) Green algae

Q12. When tuft of flagella is present at one pole of bacteria, which term we will use?

(A) Lophotrichous
(B) Amphitrichous
(C) Peritrichous
(D) Atrichous

Q13. Pigs are source of hepatitis type:

(A) B
(B) C
(C) D
(D) E

Q14. The attachment of two subunits of ribosomes is controlled by:

(A) Fe^{2+}
(B) Ca^{2+}
(C) Mg^{2+}
(D) Fe^{3+}

Q15. If non-protein part of an enzyme is loosely attached to the protein part, it is known as:

(A) Coenzyme
(B) Holoenzyme
(C) Activator
(D) Prosthetic group

Q16. How many hydrogen bonds are present in adenine and thymine pair in DNA?

(A) Two
(B) Three
(C) One
(D) Six

Q17. Which one of these is percentage of oxygen by mass of human being?

(A) 10%
(B) 65%
(C) 18%
(D) 1%

Biology (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 1115-XI124

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Define specific heat capacity of water.

(ii) Differentiate between cofactor and activator.

(iii) What is feedback inhibition of enzymes?

(iv) How substrate concentration affects enzyme activity?

(v) What is nuclear mitosis? In which organisms it is found?

(vi) Differentiate between ectomycorrhizae and endomycorrhizae.

(vii) Differentiate between Radiata and Bilateria.

(viii) What is mantle? Write its function.

(ix) Differentiate the spiral and determinate cleavage.

(x) What is Archaeopteryx? Write its reptilian and avian characters, (one each)

(xi) What is compensation point?

(xii) Define action spectrum and absorption spectrum.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) How organelle is different from organ?

(ii) What is the biological control? Give an example.

(iii) Write first two salient features of cell theory.

(iv) Define endocytosis and exocytosis.

(v) How kingdom protista was created?

(vi) Differentiate micronucleus and meganucleus in ciliates.

(vii) Compare slime molds with fungi.

(viii) Why brown algae are important?

(ix) What is the importance of photorespiration?

(x) It is said that "smokers invite Cancer". How?

(xi) How artificial pace maker works?

(xii) Differentiate antigen and antibody.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Write the scientific name of brinjal and onion.

(ii) What are super blue green algae? Give their importance.

(iii) What is meant by phylogenetic system of classification?

(iv) Why seed is considered a crucial adaptations for terrestrial life of plants?

(v) Why bryophytes are called amphibious plants?

(vi) What is meant by heterogamy?

(vii) What is botulism? Give its cause.

(viii) Can we live without large intestine? Comment.

(ix) What do you know about detritivores? Give an example.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5.

Q6.

Q7.

Q8.

Q9.

GROUP – II

Biology (Objective) Group - II

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6464

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. If ψw = -800kPa and ψs = -1400kPa, then ψp will be:

(A) 600 kPa
(B) - 600 kPa
(C) - 2200 kPa
(D) 2200 kPa

Q2. How many liters of blood are present in man whose body weight is 96kgs?

(A) 6
(B) 7
(C) 8
(D) 9

Q3. When carbon dioxide pressure increases, the capacity of haemoglobin to hold oxygen is:

(A) Decreased
(B) Increased many folds
(C) Remained constant
(D) Doubled

Q4. Dipeptides are broken down into amino acids by:

(A) Lipase
(B) Pepsin
(C) Trypsin
(D) Erypsin

Q5. The NADPH molecule reduces the sugar during in:

(A) Cyclic phosphorylation
(B) Non cyclic phosphorylation
(C) Calvin cycle
(D) Electron transport chain

Q6. The molecular formula of chlorophyll "b" is:

(A) $C_{55}H_{72}O_5N_4Mg$
(B) $C_{55}H_{70}O_6N_4Mg$
(C) $C_{50}H_{72}O_5N_4Mg$
(D) $C_{50}H_{70}O_6N_4Mg$

Q7. The left aortic arch is present in:

(A) Cat
(B) Crow
(C) Frog
(D) Cockroach

Q8. Polymorphism is the characteristics of phylum:

(A) Mollusca
(B) Arthropoda
(C) Coelenterata
(D) Porifera

Q9. The earliest group of vascular plants belongs to:

(A) Psilopsida
(B) Lycopsida
(C) Sphenopsida
(D) Pteropsida

Q10. Citric acid is obtained from a species of:

(A) Aspergillus
(B) Penicillium
(C) Saccharomyces
(D) Neurospora

Q11. Polysiphonia is an example of:

(A) Green algae
(B) Red algae
(C) Brown algae
(D) Golden algae

Q12. Which type of the bacterium E.coli is?

(A) Aerobic
(B) Anaerobic
(C) Microaerophilic
(D) Facultative anaerobic

Q13. The scientific name of tomato is:

(A) Solanum nigram
(B) Solanum tuberosum
(C) Solanum esculentum
(D) Allium cepa

Q14. Organelles found in both prokaryotic and eukaryotic cells are:

(A) Endoplasmic reticulum
(B) Mitochondria
(C) Ribosomes
(D) Lysosomes

Q15. Vitamins are the essential raw material for the synthesis of:

(A) Prosthetic group
(B) Coenzyme
(C) Activator
(D) Apoenzyme

Q16. The secondary structure of protein is found in:

(A) Trypsin
(B) Insulin
(C) Glucagon
(D) Keratin

Q17. Tentative explanation of observation is called as:

(A) Hypothesis
(B) Deduction
(C) Theory
(D) Law

Biology (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 1116-XI124

> Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) How yeast differs from other fungi?

(ii) What is parasexuality? Give its importance.

(iii) Write three general characteristics of animals.

(iv) Name two animals in which hairs have become modified.

(v) How water enters water vascular canals in echinoderm? Name that structure and its location on body side.

(vi) What are two adaptations for parasitic mode of life in flatworms?

(vii) Why do all biochemical reactions not follow the lock and key model?

(viii) What are enzymes? Give their importance.

(ix) How does enzymes accelerate the rate of metabolic reaction?

(x) Why photosynthesis is called redox process? Write its equation.

(xi) What is photosystem? Name its two parts.

(xii) How would you identify starch and glycogen solution?

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Define bioelements. Name the bioelements which occur in traces in human body.

(ii) Distinguish the micromolecules and macromolecules.

(iii) Give two functions of endoplasmic reticulum.

(iv) Define cell. Who discovered the cell?

(v) What are kelps? Give their structure.

(vi) Differentiate the diatoms and dinoflagellates.

(vii) How algae differ from plants?

(viii) Define water blooms. What is their effect on animals?

(ix) Differentiate the organismic respiration from cellular respiration.

(x) What are tracheoles in cockroach and state their function?

(xi) Name four parts of heart of fishes.

(xii) Differentiate the osmotic potential and pressure potential.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Differentiate the phage virus and a prophage.

(ii) What are plasmids? Give their importance for bacteria.

(iii) How would you clarify microphylls and megaphylls?

(iv) Why calyx and corolla are called non-essential reproductive parts of flower?

(v) How ovules of gymnosperms differs from that of angiosperms?

(vi) What are arthrophyte plants? Give example.

(vii) Write names and position of salivary glands in man.

(viii) Where are the villi located? Give their role.

(ix) How Sundew (Drosera) shows its insectivorous activity?

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5.

Q6.

Q7.

Q8.

Q9.